WORKED SOLUTION

STEP II 2005, Question 1

Pure • Polynomials and Stationary Points

STEP II 2005, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q1

The key idea

Differentiating anything of the form $P(x)\mathrm{e}^{-x^2}$ by the product rule gives $(P'(x) - 2xP(x))\mathrm{e}^{-x^2}$, and since e^{-x^2} is never zero, the stationary points are exactly the real roots of the polynomial $P'(x) - 2xP(x)$. So the second part becomes a reverse problem: decide which polynomial we want $P' - 2xP$ to be, then work backwards to P by comparing coefficients.

Solution

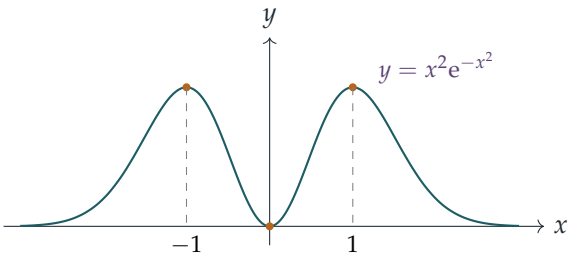
The first part

By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\mathrm{e}^{-x^2}\right) = 2x\,\mathrm{e}^{-x^2} + x^2\cdot(-2x)\,\mathrm{e}^{-x^2} = 2x(1-x^2)\mathrm{e}^{-x^2}.$$

Since $\mathrm{e}^{-x^2} > 0$ for every x, this is zero exactly when $x(1-x^2) = 0$, that is, at

$$x = 0, \qquad x = 1, \qquad x = -1.$$



The graph has a minimum at the origin and maxima of height e^{-1} at $x = \pm 1$.

The second part

For any polynomial P,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(P(x)\mathrm{e}^{-x^2}\right) = \left(P'(x) - 2xP(x)\right)\mathrm{e}^{-x^2},$$

so, again because $\mathrm{e}^{-x^2} > 0$, the derivative vanishes exactly where $P'(x) - 2xP(x) = 0$. The first part fits this pattern: $P(x) = x^2$ gave $P' - 2xP = -2x(x^2 - 1)$. So we aim for

$$P'(x) - 2xP(x) = -2x\left(x^2 - a^2\right)\left(x^2 - b^2\right), \qquad (*)$$

whose only real roots are 0, $\pm a$ and $\pm b$. These are five *distinct* values, because a and b are distinct and positive.

Ask yourself What degree should P have, and why should it contain only even powers?

If P has degree n with leading coefficient c, then P' has degree $n-1$, so $P' - 2xP$ has degree $n+1$ with leading coefficient $-2c$. The right-hand side of $(*)$ has degree 5 and leading coefficient -2, so we need $n = 4$ and $c = 1$. Moreover, the right-hand side is an odd function; if P is even, then P' and $2xP$ are both odd, so it is natural to try

$$P(x) = x^4 + \beta x^2 + \gamma.$$

Then $P'(x) = 4x^3 + 2\beta x$ and

$$\begin{aligned} P'(x) - 2xP(x) &= 4x^3 + 2\beta x - 2x^5 - 2\beta x^3 - 2\gamma x \\ &= -2x^5 + (4 - 2\beta)x^3 + (2\beta - 2\gamma)x, \end{aligned}$$

while the right-hand side of $(*)$ expands to

$$-2x^5 + 2\left(a^2 + b^2\right)x^3 - 2a^2b^2x.$$

Comparing the coefficients of x^3 and of x:

$$\begin{aligned} 4 - 2\beta = 2\left(a^2 + b^2\right) \qquad &\Longrightarrow \qquad \beta = 2 - a^2 - b^2, \\ 2\beta - 2\gamma = -2a^2b^2 \qquad &\Longrightarrow \qquad \gamma = \beta + a^2b^2 = a^2b^2 - a^2 - b^2 + 2. \end{aligned}$$

So one suitable polynomial is

$$\boxed{P(x) = x^4 + \left(2 - a^2 - b^2\right)x^2 + a^2b^2 - a^2 - b^2 + 2.}$$

With this choice $(*)$ holds identically, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(P(x)\mathrm{e}^{-x^2}\right) = -2x\left(x^2 - a^2\right)\left(x^2 - b^2\right)\mathrm{e}^{-x^2},$$

which is zero for $x = 0$, $x = \pm a$ and $x = \pm b$, and, since $\mathrm{e}^{-x^2} \neq 0$, for no other values of x. Any non-zero constant multiple of P works equally well.

Checks

- Take $a = 1$, $b = 2$: then $P(x) = x^4 - 3x^2 + 1$, and $P' - 2xP = 4x^3 - 6x - 2x^5 + 6x^3 - 2x = -2x^5 + 10x^3 - 8x = -2x(x^2 - 1)(x^2 - 4)$, as required.
- Numerically, for $a = 0.7$, $b = 1.9$, the derivative of $P(x)\mathrm{e}^{-x^2}$ changes sign only near $x = 0, \pm 0.7, \pm 1.9$ on $[-5, 5]$; the formula was also confirmed at thousands of random (a, b, x).
- Letting $b \to 0$ heuristically: $P \to x^4 + (2 - a^2)x^2 + 2 - a^2$, and the derivative becomes $-2x^3(x^2 - a^2)\mathrm{e}^{-x^2}$, with the roots at 0 and $\pm b$ merging.

Common slips

- Forgetting the chain-rule factor: the derivative of e^{-x^2} is $-2x\,\mathrm{e}^{-x^2}$, not e^{-x^2}.
- Guessing P of degree 5 to match the five roots: differentiating $P\mathrm{e}^{-x^2}$ *raises* the degree by one, so P has degree 4.
- Not saying why there are no other stationary points: this needs $\mathrm{e}^{-x^2} \neq 0$ and the fact that the quintic has only these roots.

Beyond STEP

The map $P \mapsto P' - 2xP$ is a linear operator on polynomials, and solving $(*)$ is solving a first-order linear differential equation with an integrating factor e^{-x^2}. Functions of the form $H(x)\mathrm{e}^{-x^2}$ with polynomial H lead to the Hermite polynomials, which appear in probability (the normal distribution) and in the quantum harmonic oscillator. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors and Hermite's equation.

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